

conquer a lion's share of this lucrative market.

How to install Asterisk as an IP PBX?

A step-by-step tour of how to install and configure Asterisk as an IP PBX for a 100-people organisation would help you to understand the complexities involved and flexibilities available in this technology.

Hardware selection. It starts with selection of hardware to install the IP PBX. Normally, for a 100-employee organisation, a standard P IV machine, 80GB hard disk and 2GB RAM are enough. The hardware should have a backplane suitable to accommodate the trunk interfaces and any FXS card selected below. Normally, all of the trunking and subscriber hardware supports PCI bus. Therefore it is good if the motherboard has a few free PCI slots available.

Software selection. There are various versions of Asterisk available: Asterisk original version (www.asterisk.org), AsteriskNow (www.asterisknow.org) and TrixBox (www.trixbox.com). Matching your requirements to the features provided by each version will help you to decide the variant. You may choose AsteriskNow or TrixBox if you intend to install a simple IP PBX. For complex applications like call centres and prepaid calling card platforms, you may choose the original version and install the respective application.

Trunk interfaces. Normally, you need to connect the IP PBX to the PSTN (the ordinary public switched telephone network we use). For this, you need additional hardware installed in your system. FXO interfaces help you to connect with normal analogue trunks. ISDN PRI interfaces are quite common for installations above 50 users, as these provide a more reliable digital connection. There are also GSM cards available for users to connect directly to the mobile network and thereby save huge telephone costs.

Digium offers a variety of hardware to support analogue trunks, dig-

Asterisk, with its flexibility to be deployed as a hybrid PBX as well as pure IP PBX, is well positioned to conquer a lion's share of the PBX market

ital BRI, PRI, etc. There are also other hardware manufacturers like Sangoma (www.sangoma.com) and ZapMicro (www.zapmicro.com) whose products you may like to explore.

Phones and ATAs. Finally, you need to connect the users to the IP PBX, so that they can make calls. Four main choices available are IP phones, dual phones, softphones, FXS ports and ATAs.

IP phones come in various forms from multiple manufacturers like Cisco, SNOM and Grandstream.

Examples of dual phones are models like Nokia E60, which supports GSM and WLAN. These phones can be used for GSM network outside the office and inside the office they can work as IP PBX extensions.

Softphones (like X-Lite, eyebeam and SJ Phone) are available in plenty on the Internet. These can be installed on the desktops and used together with headsets/microphones.

The IP PBX can also be installed with additional hardware with FXS ports, so you can connect normal analogue phones to these ports.

Installation. Installation involves server installation, trunk installation and network/terminal equipment installation.

Server installation starts with installing Linux on the server hardware—it could be Redhat, Fedora, CentOS, etc. Further, the Asterisk version should be downloaded and installed using the standard instructions available with the variant. A good knowledge of Linux will be very helpful to carry out these steps smoothly.

Trunk installation involves mechanical installation of the selected

hardware in the server, installation of the driver software and then configuration of the driver.

Network installation is quite simple as it can use the existing LAN/WLAN infrastructure. In case the LAN/WLAN infrastructure is non-existent, this could be entrusted to a network service company. Further, the phones need to be configured with the corresponding extension number, registrar IP, etc.

World beyond IP PBX

Asterisk is not only an IP PBX, but it is a platform that can host multiple telephony applications like:

1. **Transit switching.** It can act as a Class 4 switch to switch between various suppliers and carriers for international traffic.

2. **IVR.** It can act as an interactive voice response (IVR) platform and handle complex IVR call flows in call centres, info centres, etc.

3. **Call centre.** External applications like Vicidial and DialPrix are available, which can run on the platform and transform into powerful call centre applications for inbound and outbound modes. Also, applications for voice logging, monitoring, etc come at no additional cost.

4. **Calling card.** Prepaid calling cards are going to become more popular in the Indian scenario. Powerful applications like asteriskbilling and a2Billing transform Asterisk into a prepaid calling card platform.

The possibilities opened up by the Asterisk platform are endless. And anybody associated with telecom cannot ignore it in the future. A study of the platform could bring in many benefits to your organisation, by way of organising the telecom infrastructure, finding new applications in your area of work and generating a new revenue stream, or start providing services on this platform. So join the revolution before it is too late! ●

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